

4. The brown rat (*Rattus norvegicus*) spreads disease. Warfarin is a poison used to kill brown rats. Over time, many brown rats have evolved to become resistant to warfarin.



(a) State the genus of the brown rat. [1]

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(b) The brown rat is a vertebrate. State what is meant by the term vertebrate. [1]

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(c) The sentences below describe how brown rats evolved to become warfarin resistant. They are not in the correct order:

- A. the mutation made the brown rat resistant to warfarin
- B. the result is an increase in the population of brown rats resistant to warfarin
- C. a mutation occurred in a gene
- D. brown rats that were resistant to warfarin lived long enough to reproduce
- E. during reproduction the useful gene is passed on to the offspring

Use letters **A–E** from the list above to complete the sequence below to describe how brown rats became warfarin resistant. [2]

..... **C** **A**
.....

(d) State the term used when all individuals of a species have died out. [1]

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